MA104 · DISCRETE MATHEMATICS · LECTURE 2

Membership Tables

$A \triangle B$

Fifteen worked examples — set operations and set identities

How to read a membership table

Each row is one possible situation for an element x; each column records whether x belongs to the set in its heading: **1** means x is in the set, **0** means it is not. Every set operation is then a rule on bits, exactly like a truth table.

Operation	Entry is 1 when ...	Logic
$A \cup B$	x is in A **or** in B (or both)	$\vee$
$A \cap B$	x is in A **and** in B	$\wedge$
A^c	x is **not** in A	$\neg$
$A - B$	x is in A and **not** in B	$p \wedge \neg q$
$A \triangle B$	x is in **exactly one** of A, B	$\oplus$

Rule. To prove two sets are equal, build a column for each side. If the two final columns agree on **every** row, the sets are equal; one disagreeing row is a counterexample.

Basic operations on concrete sets

Example 1 Union

$$A = \{1, 2, 4\}, \quad B = \{2, 3, 4\}$$

Find $A \cup B$ using a membership table.

Element	A	B	$A \cup B$
1	**1**	0	**1**
2	**1**	**1**	**1**
3	0	**1**	**1**
4	**1**	**1**	**1**

$$A \cup B = \{1, 2, 3, 4\}$$

Example 2 Intersection

$$A = \{1, 2, 3, 5\}, \quad B = \{2, 3, 4, 6\}$$

Find $A \cap B$ using a membership table.

Element	A	B	$A \cap B$
1	**1**	0	0
2	**1**	**1**	**1**
3	**1**	**1**	**1**
4	0	**1**	0
5	**1**	0	0
6	0	**1**	0

$$A \cap B = \{2, 3\}$$

Example 3 Complement

$$U = \{1, 2, 3, 4, 5, 6\}, \quad A = \{1, 3, 5\}$$

Find A^c using a membership table.

Element	A	A^c
1	**1**	0
2	0	**1**
3	**1**	0
4	0	**1**
5	**1**	0
6	0	**1**

$$A^c = \{2, 4, 6\}$$

Example 4 Difference

$$A = \{1, 2, 3, 4\}, \quad B = \{3, 4, 5, 6\}$$

Find $A - B$ using a membership table.

Element	A	B	$A - B$
1	**1**	0	**1**
2	**1**	0	**1**
3	**1**	**1**	0
4	**1**	**1**	0
5	0	**1**	0
6	0	**1**	0

$$A - B = \{1, 2\}$$

Difference and symmetric difference

Example 5 Difference Identity

Use a membership table to show

$$A - B = A \cap B^c$$

A	B	$A - B$	B^c	$A \cap B^c$
1	1	0	0	0
1	0	1	1	1
0	1	0	0	0
0	0	0	1	0

✓ **The shaded columns are identical.** Hence the identity holds for every possible membership.

Example 6 Symmetric Difference

$$A = \{1, 2, 3, 4\}, \quad B = \{3, 4, 5, 6\}$$

The common elements are $A \cap B = \{3, 4\}$. They belong to **both** sets, so they are excluded; every remaining element belongs to exactly one set:

$$A \triangle B = \{1, 2, 5, 6\}$$

Equivalent form. $A \triangle B = (A - B) \cup (B - A)$, and indeed

$$A - B = \{1, 2\}, \quad B - A = \{5, 6\}, \quad A \triangle B = \{1, 2\} \cup \{5, 6\} = \{1, 2, 5, 6\}$$

Membership table. An element is in $A \triangle B$ exactly when its two bits differ:

A	B	$A \triangle B$
1	1	0
1	0	1
0	1	1
0	0	0

Note. Symmetric difference = elements in A **or** B, but **not in both** — the set version of exclusive or ($\oplus$).

De Morgan's laws

Example 7 De Morgan's Law

Use a membership table to show

$$(A \cup B)^c = A^c \cap B^c$$

A	B	$A \cup B$	$(A \cup B)^c$	A^c	B^c	$A^c \cap B^c$
1	1	1	0	0	0	0
1	0	1	0	0	1	0
0	1	1	0	1	0	0
0	0	0	1	1	1	1

✓ **The shaded columns are identical.** Hence the identity holds for every possible membership.

Example 8 Second De Morgan's Law

Use a membership table to show

$$(A \cap B)^c = A^c \cup B^c$$

A	B	$A \cap B$	$(A \cap B)^c$	A^c	B^c	$A^c \cup B^c$
1	1	1	0	0	0	0
1	0	0	1	0	1	1
0	1	0	1	1	0	1
0	0	0	1	1	1	1

✓ **The shaded columns are identical.** Hence the identity holds for every possible membership.

Distributive and absorption laws

Example 9 Distributive Law

Use a membership table to show

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

A	B	C	$B \cup C$	$A \cap (B \cup C)$	$A \cap B$	$A \cap C$	$(A \cap B) \cup (A \cap C)$
1	1	1	1	1	1	1	1
1	1	0	1	1	1	0	1
1	0	1	1	1	0	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

✓ **The shaded columns are identical.** Hence the identity holds for every possible membership.

Example 10 Absorption Law

Use a membership table to show

$$A \cup (A \cap B) = A$$

A	B	$A \cap B$	$A \cup (A \cap B)$	A
1	1	1	1	1
1	0	0	1	1
0	1	0	0	0
0	0	0	0	0

✓ **The shaded columns are identical.** Hence the identity holds for every possible membership.

Example 11 Another Absorption Law

Use a membership table to show

$$A \cap (A \cup B) = A$$

A	B	$A \cup B$	$A \cap (A \cup B)$	A
1	1	1	1	1
1	0	1	1	1
0	1	1	0	0
0	0	0	0	0

✓ **The shaded columns are identical.** Hence the identity holds for every possible membership.

Difference laws

Example 12 Difference of a Union

Use a membership table to show

$$A - (B \cup C) = (A - B) \cap (A - C)$$

A	B	C	$B \cup C$	$A - (B \cup C)$	$A - B$	$A - C$	$(A - B) \cap (A - C)$
1	1	1	1	0	0	0	0
1	1	0	1	0	0	1	0
1	0	1	1	0	1	0	0
1	0	0	0	1	1	1	1
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

✓ **The shaded columns are identical.** Hence the identity holds for every possible membership.

Example 13 Difference of an Intersection

Use a membership table to show

$$A - (B \cap C) = (A - B) \cup (A - C)$$

A	B	C	$B \cap C$	$A - (B \cap C)$	$A - B$	$A - C$	$(A - B) \cup (A - C)$
1	1	1	1	0	0	0	0
1	1	0	0	1	0	1	1
1	0	1	0	1	1	0	1
1	0	0	0	1	1	1	1
0	1	1	1	0	0	0	0
0	1	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	0

✓ **The shaded columns are identical.** Hence the identity holds for every possible membership.

Note. Examples 12 and 13 are De Morgan's laws in disguise: "outside $B \cup C$" means "outside B **and** outside C", while "outside $B \cap C$" means "outside B **or** outside C".

Associative laws

Example 14 Associative Law of Union

Use a membership table to show

$$(A \cup B) \cup C = A \cup (B \cup C)$$

A	B	C	$A \cup B$	$(A \cup B) \cup C$	$B \cup C$	$A \cup (B \cup C)$
1	1	1	1	1	1	1
1	1	0	1	1	1	1
1	0	1	1	1	1	1
1	0	0	1	1	0	1
0	1	1	1	1	1	1
0	1	0	1	1	1	1
0	0	1	0	1	1	1
0	0	0	0	0	0	0

✓ **The shaded columns are identical.** Hence the identity holds for every possible membership.

Example 15 Associative Law of Intersection

Use a membership table to show

$$(A \cap B) \cap C = A \cap (B \cap C)$$

A	B	C	$A \cap B$	$(A \cap B) \cap C$	$B \cap C$	$A \cap (B \cap C)$
1	1	1	1	1	1	1
1	1	0	1	0	0	0
1	0	1	0	0	0	0
1	0	0	0	0	0	0
0	1	1	0	0	1	0
0	1	0	0	0	0	0
0	0	1	0	0	0	0
0	0	0	0	0	0	0

✓ **The shaded columns are identical.** Hence the identity holds for every possible membership.

Rule. To prove two sets are equal using a membership table, construct the columns for both sides. If the final two columns are identical, the two sets are equal.